

Exercise 07
Tracking MPC and LQR

Exercise 1 **MPC Tracking**

Consider the following system:

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k), \\y(k) &= Cx(k), \\A &= \begin{bmatrix} 7 & 2 \\ -2 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0.5 & 0.5 \\ 0 & 1 \end{bmatrix},\end{aligned}$$

where $x(k) \in \mathbb{R}^2$, $u(k) \in \mathbb{R}$ and $y(k) \in \mathbb{R}^2$. The following optimization problem defines a reference-tracking MPC controller of horizon two:

$$\begin{aligned}\min_U \quad & P \|r - Cx_2\|_\infty + \sum_{i=0}^1 Q \|r - Cx_i\|_\infty + R(u_s - u_i)^2 \\ \text{s.t.} \quad & x_0 = x(0), \\ & x_{i+1} = Ax_i + Bu_i, \quad i = 0, 1 \\ & u_{\min} \leq u_i \leq u_{\max}, \quad i = 0, 1,\end{aligned}$$

where $U = [u_0 \ u_1]^\top \in \mathbb{R}^2$ and r is the reference for the output $y(k)$. The quantities $u_{\min} \in \mathbb{R}$, $u_{\max} \in \mathbb{R}$, $r \in \mathbb{R}^2$, $R > 0$, $Q \geq 0$ and $P \geq 0$ are given with $R, Q, P \in \mathbb{R}$.

- 1) Is the problem convex? Justify your answer.
- 2) Given the output reference value $r = [2 \ 3]^\top$, find the steady state x_s that leads to this output as well as the corresponding steady state input u_s .
- 3) Reformulate the MPC tracking problem with the delta-formulation.

Exercise 2 **Tracking LQR**

Consider the unconstrained linear time-invariant system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k)\end{aligned}$$

- 1) Show how to compute an LQR control law $u(x, r)$ that will cause the output of the system y to track a given constant reference r .
 - (i) State an expression to compute a steady-state (x_s, u_s) such that the output of the system is equal to a desired reference r .
 - (ii) State the infinite-horizon LQR problem for the state and input weighting matrices Q and R in delta-form.
 - (iii) State the control law $u(x, r)$.
- 2) Consider the system $x(k+1) = 2x(k) + u(k)$, $y(k) = x(k)$ and design a tracking LQR controller for the weights $Q = R = 1$. Give the control law $u(x, r)$.